

September dataset and experiment update

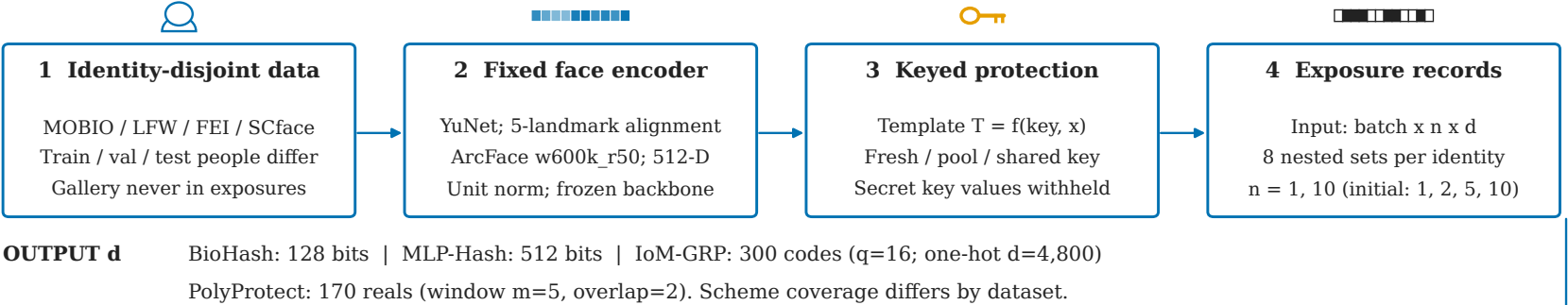
Four multi-exposure datasets; FEI and SCface complete the two-added-dataset target.

Dataset	Identities	Train / val / test	Valid / selected	Variation
MOBIO	150	90 / 30 / 30	1,799 / 1,800	Sessions
LFW	125	75 / 25 / 25	1,500 / 1,500	In-the-wild
FEI	200	120 / 40 / 40	2,378 / 2,400	Pose / expression
SCface	130	78 / 26 / 26	2,851 / 2,860	Camera / distance

SCface access supplied by Sani on 2026-09-18; acquisition is complete.
Mugshot gallery / surveillance exposures; all 130 identities remain eligible after 9 detection failures.
Unprotected SCface top-1: 84.375% over 544 test probes; chance: $1/26 = 3.846\%$.
Evidence: 73 key-pool conditions / 12 studies; 72 one-seed model endpoints / 24 scheme pilot cells.
AgeDB is optional and not acquired. Multi-seed scheme confirmation remains open.

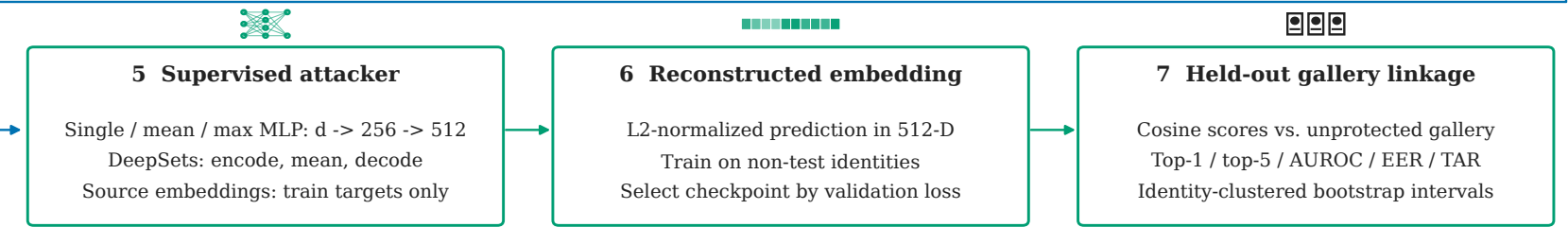
Experimental architecture

A DATA AND PROTECTION



B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



KEY DESIGN Fresh: source-record keys are split-disjoint. Pool K: hidden transforms recur across identity splits.

UNCERTAINTY Earlier studies: 3 model seeds. IoM-GRP / PolyProtect pilots: 1 seed; not confirmation.

SCOPE Tests transform reuse and same-identity aggregation, not a new backbone or a universal privacy guarantee.

Icons are schematic; no biometric images or secret keys are included. Detailed model definitions follow.

Model definitions and evaluation protocol

INPUTS AND MODELS

T: batch $\times$ n $\times$ d. Single MLP: $d \rightarrow 256 \rightarrow \text{ReLU} \rightarrow 512 \rightarrow \text{L2 normalization}$.

Mean / max MLP: pool records first, then the same MLP. Key-pool endpoint: mean MLP.

DeepSets phi: $d \rightarrow 256 \rightarrow \text{ReLU} \rightarrow 256 \rightarrow \text{ReLU}$; masked mean across records.

DeepSets rho: $256 \rightarrow 256 \rightarrow \text{ReLU} \rightarrow 512 \rightarrow \text{L2 normalization}$; permutation-invariant.

BioHash: 128 bits. MLP-Hash: 512 bits. IoM-GRP: 300 codes, $q=16$, one-hot $d=4,800$.

PolyProtect: 170 real values; window 5, overlap 2; outside the rotational-invariance theorem.

TRAINING AND EVALUATION

Paired source embeddings supervise training; loss = $\text{mean}(1 - \text{cosine}) + 0.1 \times \text{MSE}$.

Adam: learning rate 0.001, weight decay 0.0001; best validation-loss checkpoint.

SCface BioHash: 3 seeds, 400-epoch cap, patience 60. Scheme pilots: 1 seed, 120 / 30.

Eight nested sets per identity; gallery image excluded. Test identities never used for training.

Fresh keys: source-record and split-disjoint. Recurring pools deliberately reuse hidden transforms.

Cosine gallery linkage; top-1/top-5, AUROC, EER, TAR; 2,000 identity-bootstrap resamples.

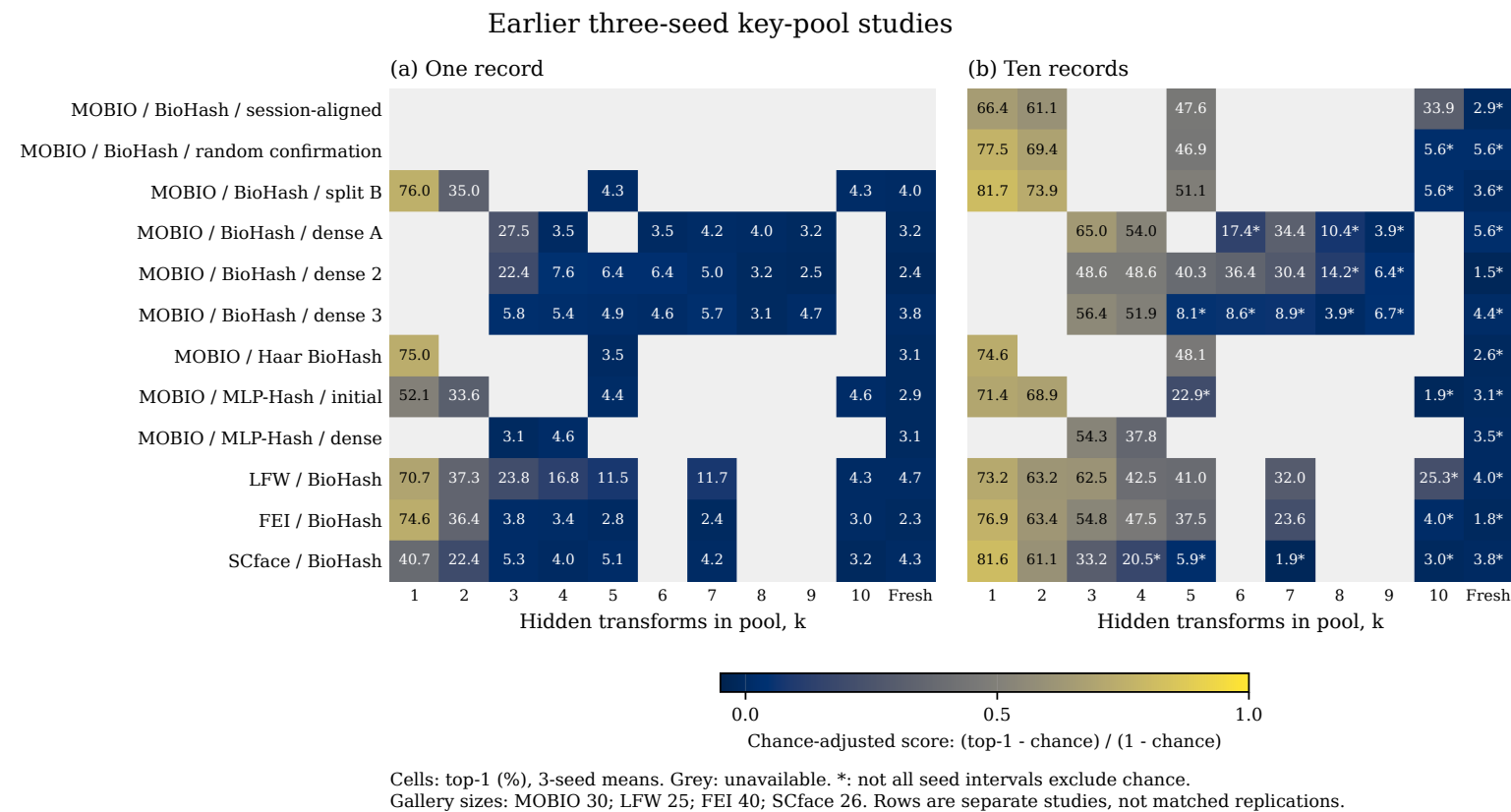
95% intervals condition on the identity partition/model seed; no multiplicity adjustment.

Sources: `src/biometrics_ai/aggregation/models.py`; `scripts/train/run_real_multiexposure.py`; `configs/attacks/scface*.yaml`

Cross-dataset key-reuse boundary

SCface: pool 3 rises from 5.29% to 33.17% top-1; fresh keys give 3.85%.

SCface pools 1/2/3 pass the all-seed interval criterion; 4/5/7/10 fail. Failure is not equivalence.

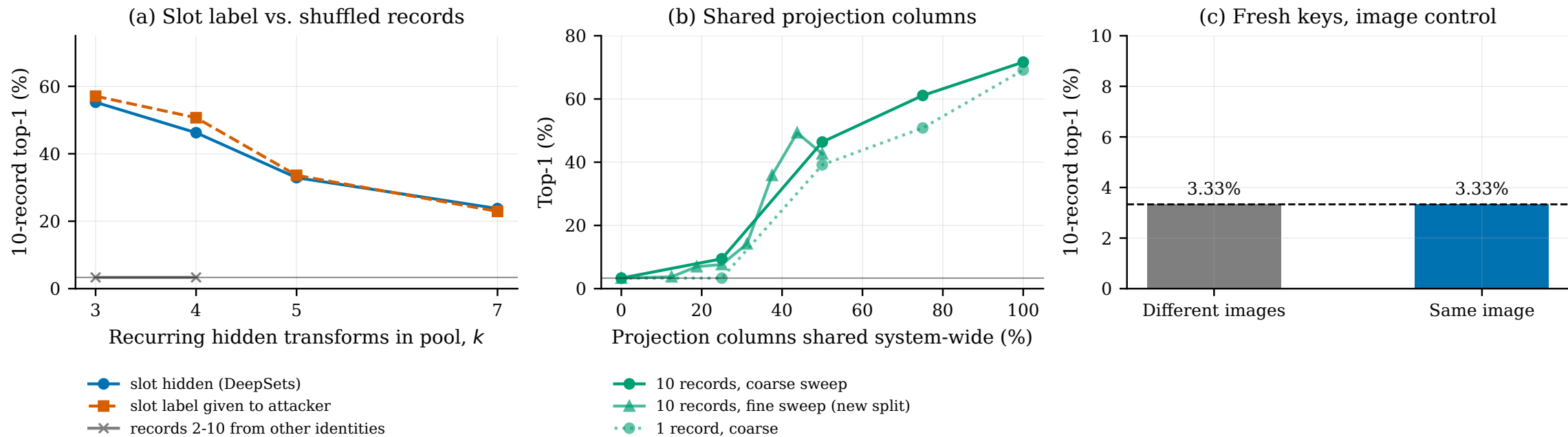


73 conditions / 12 source-separated, three-seed studies. Grey: unavailable. *: interval failure. Pilots excluded.

Mechanism controls and interpretation

MOBIO: slot labels change DeepSets by -0.83 to +4.44 points; shuffled records collapse to 3.33% chance.
Same-image fresh-key sets also give 3.33%. Projection-sharing effects depend on construction and partition.

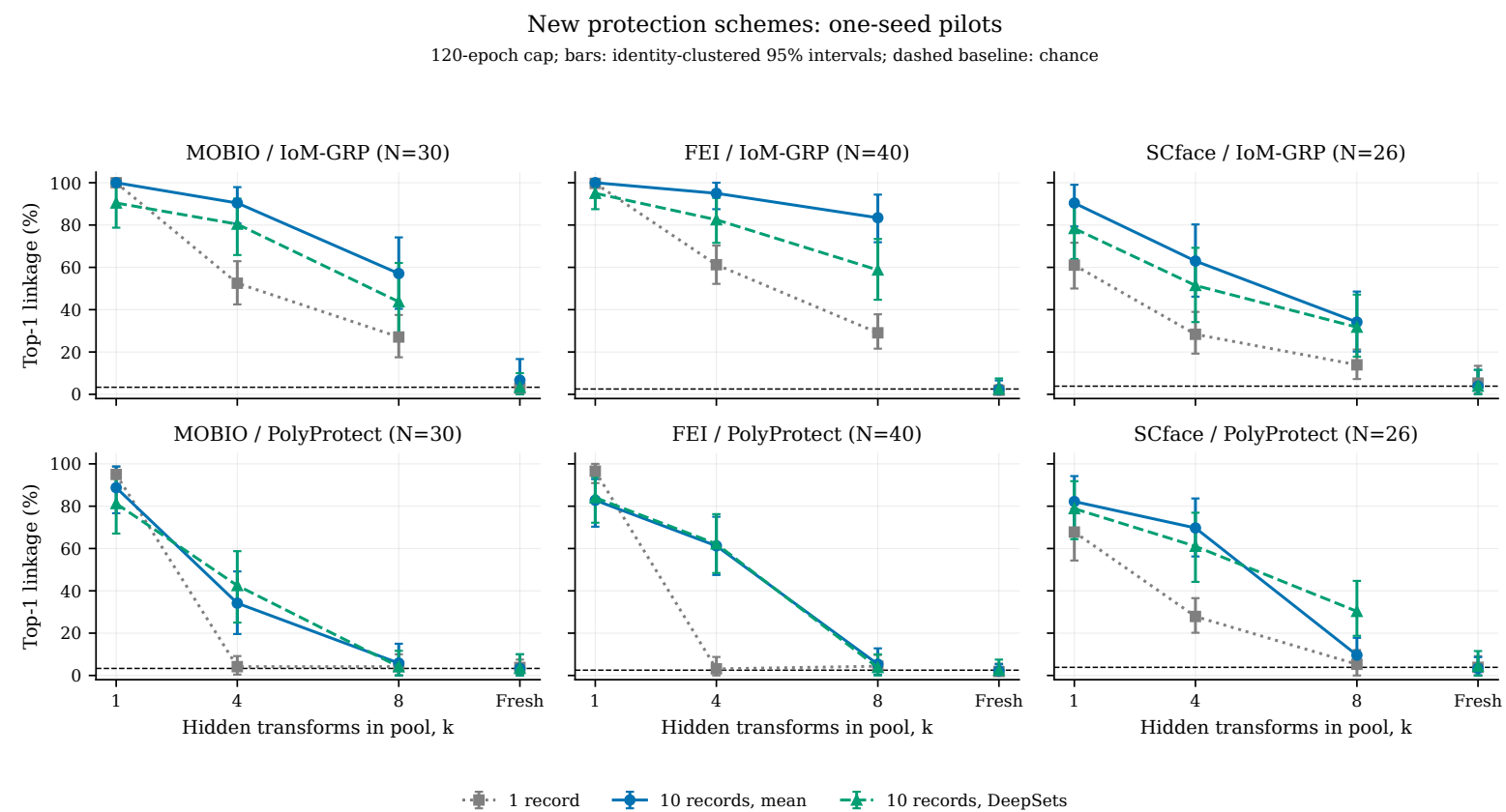
MOBIO mechanism controls (30 gallery identities; chance 3.33%)



Sources: experiments/mobio_mechanism_controls and mobio_correlation_controls. Three model seeds; chance 1/30.

Additional schemes: one-seed pilots

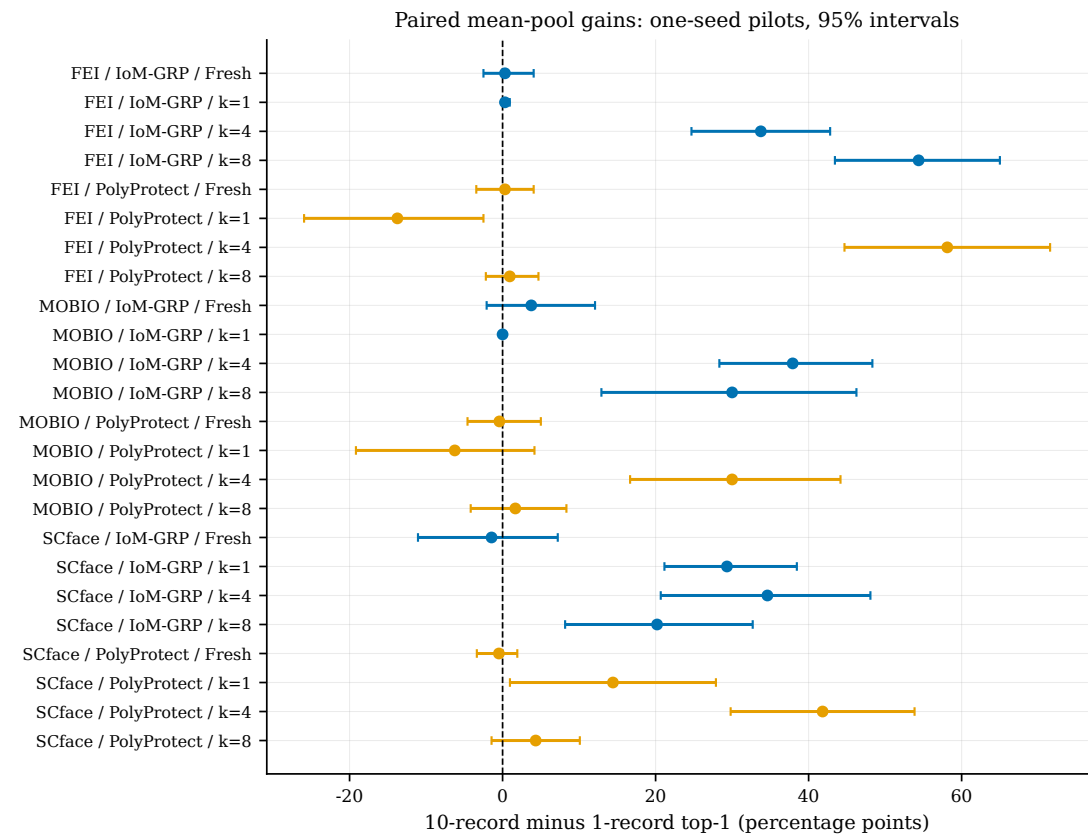
72 model endpoints: MOBIO / FEI / SCface; IoM-GRP / PolyProtect; pools 1/4/8 and fresh keys.



One seed per endpoint, 120-epoch cap. No controlled ranking against earlier three-seed, 400-epoch studies.

Paired multi-record gains and uncertainty

SCface pool-4 mean-pool gains: IoM-GRP +34.62 points; PolyProtect +41.83 points.

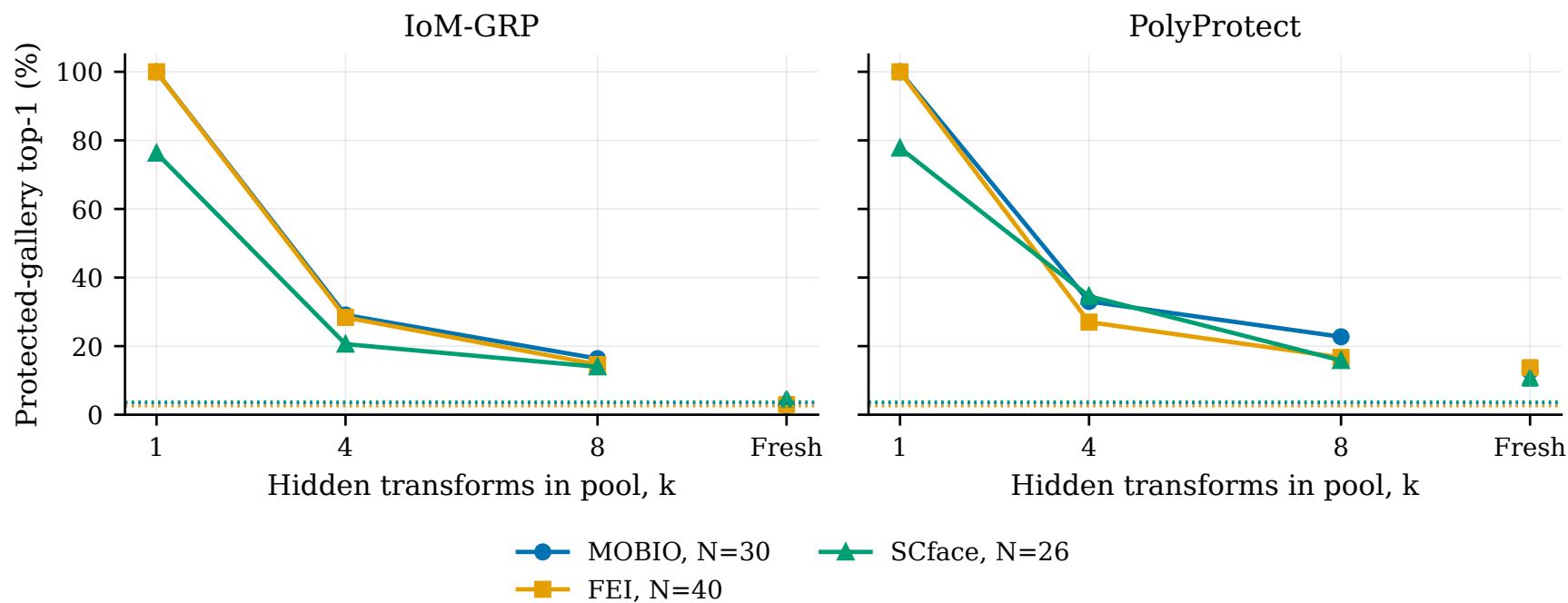


Paired 95% identity-bootstrap intervals, conditional on one seed/partition; unadjusted. These are pilots, not confirmation.

Native matching is a separate diagnostic

Fresh-key PolyProtect native top-1: 12.73% / 13.73% / 10.66% on MOBIO / FEI / SCface.
Above-chance native identification needs investigation; learned attacks near chance do not imply unlinkability.

Native matching diagnostic, not learned embedding linkage



Protected-gallery diagnostic with different probes; not the learned unprotected-gallery endpoint. No universal privacy claim.

Evidence provenance and submission gates

PROVENANCE

Latest result integration: 4352eeb (2026-09-18). SCface protocol freeze: 69a93e4.

MOBIO / FEI scheme pilot freeze: d5f4e89 (2026-09-12). Original run dates remain unchanged.

Sources: experiments/cross_dataset_key_pool_summary.csv and both scheme pilot directories.

Local per-seed inventory: 633 rows / 25 artifacts; SCface absent locally.

All 12 current vector figures: reports/slides/figure_appendix.pdf; regenerators: scripts/figures/.

REQUIRED BEFORE SUBMISSION

1. Authorize and freeze multi-seed, multi-partition confirmation with matched training budgets.
2. Approve equivalence margins and a seed-uncertainty / multiple-comparison plan.
3. Investigate native PolyProtect linkage and norm-sensitive controls; reconcile historical coverage.
4. Obtain independent novelty, theorem and implementation review, then Sani's scientific review.
5. Select a venue and check its reporting, reproducibility, ethics and dataset-use requirements.

A/A* is a venue ambition, not a verified quality label or acceptance prediction.

No source-exact benchmark reproduction, confirmation, or universal privacy result is claimed.